Lesson 42: Logos

English Stage 3 (C1) · Unit 5: Argument and Academic Style

Adapted from Reading, Writing, Research, and Reasoning: An Advanced ESL Text by Gabriel Winer and Elizabeth Wadell, ASCCC Open Educational Resources Initiative (CC BY-NC 4.0): 5.6 Identifying and Using Logos.

In this lesson

You will study logos, the appeal to logic, and describe the logical "bones" of an argument as a syllogism: a major premise, a minor premise and a conclusion. You will analyze the logic of a short article about consumer psychology research and plan the logic of your own argument. This helps you judge whether a research article really shows what it claims, and write essays in which every reason supports the thesis.

Learning goals

After this lesson you should be able to:

- Explain what an appeal to logos is and test the logos of a text.
- Identify the major premise, minor premise and conclusion of an argument.
- Explain the difference between a valid argument and a true one.
- Complete and build syllogisms for an argumentative essay.
- Plan an introduction and topic sentences that state your common ground, thesis and reasons.

Key language

Word or phrase	Meaning	Example
logos	an appeal to logic: a structure of reasons supported by evidence	To use logos is to point out facts we can measure, count, and agree on.
build a case (for)	present reasons and evidence step by step, as a lawyer does	We sometimes call this "building a case" for our position.
common ground	a position that everyone can already agree with	We all want a cleaner environment.
categorical syllogism	a logical argument in three parts: two premises and a conclusion	Petroleum products cause pollution. Polyester fabric is made from petroleum. Therefore, polyester fabric causes pollution.
PREM·ise	a statement that an argument is based on	If both premises are true, the conclusion has to be true.
major premise	a general idea or assumption, often about values, that any reasonable person accepts	Toxic chemicals should be kept out of the water.

Word or phrase	Meaning	Example
minor premise	a more specific claim that you show to be true with evidence	Textile factories in a particular town are releasing ammonia and nitrates into local rivers.
conclusion (of a syllogism)	what must be true if both premises are true; in an essay, the thesis	Therefore, those factories should find a way to process their toxic waste without dumping it into the water.
valid	(of an argument) the conclusion must follow from the premises	A syllogism can be valid without being true.
deductive logic	reasoning from a general idea to a specific conclusion	Identify the deductive logic in the form of a syllogism.
hy·POTH·e·sis	a prediction that a study tests	To test our hypothesis, we studied how well 236 undergraduates would remember…
participants	the people who take part in a study	…participants who saw the child-labor jeans said they felt more uncomfortable.
significantly (in research)	by a difference large enough that it is unlikely to be due to chance	The participants were significantly less likely to correctly remember when a desk was made with rainforest wood.

Rules and explanations

What is logos?

An appeal to logos is an appeal to logic. The writer points out facts that we can measure, count and agree on, supports each reason with evidence, and then argues that because we agree on these facts, we must also agree on ideas that are bigger than the facts.

Reading for logos. Take the ideas apart as if they were a math problem:

- Does the writer really support each reason with strong evidence?
- Do the reasons work together to support the writer's main idea?
- Can we find any places where the logic falls apart?

Writing with logos. Build a case for your thesis like a lawyer:

1. Start from **common ground**: a position everyone can agree with (or would pretend to agree with, so that they seem like a good person). We all want a cleaner environment, even if we disagree about how to make it cleaner.
2. Write a strong thesis that is **narrow enough to defend**.
3. Write clear **topic sentences** that state reasons your thesis is true, name the areas of the thesis each paragraph explains, or set up concessions and counterarguments (Lesson 44).
4. Use clear **connecting words** to show how ideas are related, which ideas are more specific, and where you change perspectives.

The categorical syllogism

If we all agree on one big idea, and you show that a smaller idea is true, then we have to agree on a third idea.

Part	What it is	Where it goes in an essay	Aristotle's example
major premise	a general idea or assumption, often about values: the common ground	introduction (between the hook and the thesis), conclusion paragraph, and/or explanation after evidence	All humans are mortal (do not live forever).
minor premise	a more specific claim you show to be true with evidence	topic sentences	All Greeks are humans.
conclusion	must be true if both premises are true	the thesis	All Greeks are mortal.

- To find a major premise, ask: What should be true? What do we all agree is right or wrong? What do we all want for our families or societies?
- State the major premise even if it seems obvious: it reminds readers that they agree with the foundation of your argument.
- The terms describe the logic underneath your words, not the order of your essay. The "conclusion" of a syllogism is your thesis, not your conclusion paragraph.
- A syllogism can lead to a fact (polyester causes pollution) or to an action (the factories should stop dumping toxic waste into rivers).

Valid vs. true

- **Valid:** if the two premises are true, the conclusion must be true.
- **True:** the premises really are true.
- An argument can be valid without being true, and it can contain true statements without being valid (Exercise 1). The logic of an argumentative essay should be both valid and true.

> **Note for Arabic speakers** - Arabic persuasive writing often uses repetition and emphatic statements to persuade. English academic readers expect structure instead: each reason needs its own evidence, and the links between ideas need explicit connecting words such as because, therefore and as a result.

Practice

Exercise 1: Valid, true, or both? For each syllogism, decide: (a) Is it valid? (b) Are its premises true?

1. All humans are mortal. All Greeks are humans. All Greeks are mortal.
2. All people who wear glasses have X-ray vision. Ahmed wears glasses. Ahmed has X-ray vision.
3. Anyone who lives in Sacramento lives in California. Esme does not live in Sacramento. Therefore, Esme does not live in California.

Exercise 2: Practice with syllogisms. Finish the incomplete part in each set to make a valid syllogism.

1. **Major premise:** Governments are responsible for regulating fair wages. **Minor premise:** The workers in the textile factories in Bangladesh that export clothing to the U.S. are not earning a living wage. **Conclusion:** Therefore, the U.S. and Bangladeshi governments should step in to _____.
2. **Major premise:** To help change worker exploitation, we should use the strategy the workers in the situation are asking for. **Minor premise:** In this case, the workers do not want U.S. consumers to _____ the company. **Conclusion:** Therefore, if we want to help, we should use a different strategy besides boycotting.
3. **Major premise:** We can all agree that _____ is more important than temporary profits. **Minor premise:** Retrofitting factory exhaust machinery costs money now, but reduces air pollution for years. **Conclusion:** Factory owners should spend the money on retrofitting the exhaust machinery.

Exercise 3: Finding deductive logic in a text. Read "Untrustworthy Memories Make it Hard to Shop Ethically" in the lesson material. Pay attention to the logical structure of the whole article. What is the major premise? What is the minor premise? What is the logical conclusion?

Exercise 4: Reading the evidence. Answer these questions about the same article.

1. What did the researchers predict (their hypothesis)?
2. For each of the three studies, give the participants, what they read or studied, and the main result.
3. Why did the researchers also test how well the students remembered other attributes of the desks, such as their prices?
4. "We determined that this desire to not feel uncomfortable again led participants to forget about the child labor detail." Did the study measure this directly, or is it an interpretation?

Speak with your tutor

- "Here is the syllogism under my essay: [major premise, minor premise, conclusion]. Is it valid? Are both premises true? Which premise is the weakest link?"
- "Let's debate. My thesis is: [for example, 'Shoppers who care about ethics should use a written guide instead of relying on memory']. Take the opposite side and attack my premises. After five turns, tell me which of my reasons were strongest, whether I supported each with evidence, and whether my connecting words made the logic clear."
- "Give me three incomplete syllogisms about topics from my psychology course, such as memory, sleep or stress. I will complete them; then tell me whether each one is valid and true."
- "Ask me five questions about the studies in 'Untrustworthy Memories Make it Hard to Shop Ethically'. Correct my answers and my use of participants, hypothesis and significantly."

Read aloud

Pause at each |.

1. Petroleum products cause pollution. | Polyester fabric is made from petroleum. | Therefore, | polyester fabric causes pollution.
2. A syllogism can be valid | without being true.
3. The logic that forms the "bones" of an argumentative essay | should be both valid and true.
4. Our research shows that | even though most consumers want to buy ethically sourced items, | it's hard for them to heed these sentiments.
5. To test our hy·POTH·e·sis, | we studied how well 236 (two hundred and thirty-six) undergraduates | would remember manufacturing information about six wooden desks.
6. The key is to realize our memories are not perfect | and that shopping without a plan | may lead us away from our values.

Writing task

Plan an argument with strong logos and write its introduction.

1. Choose a topic: one of the book's (ethical shopping and child labor, factory pollution, fair wages in the garment industry) or a psychology topic, for example "Students should sleep instead of studying all night before an exam" (sleep and memory) or "Shoppers who want to buy ethically should not rely on their memory".
2. Write your syllogism: a major premise (common ground), one to three minor premises (reasons you can support with evidence), and a conclusion (your thesis).
3. Write an introduction of 120-180 words: a hook, the major premise as common ground, then your thesis.
4. Write one topic sentence (a minor premise) for each body paragraph, and note the evidence you will use. For psychology topics, look in the free OpenStax "Psychology 2e" textbook or on Google Scholar.

Checklist:

- Would any reasonable reader accept your major premise?
- Can you show each minor premise to be true with evidence?
- If both premises are true, must your thesis be true (is it valid)?
- Are your premises really true (no "X-ray vision" premises)?
- Is your thesis narrow enough to defend?
- Is the major premise between the hook and the thesis?
- Does each topic sentence state one reason?
- Do connecting words (because, therefore, as a result) show how your ideas are linked?

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- The original article "Untrustworthy Memories Make it Hard to Shop Ethically" on The Conversation, with links to the research it reports: https://theconversation.com/untrustworthy-memories-make-it-hard-to-shop-ethically-87572

Answer key

Exercise 1.

1. (a) Valid. (b) True: Aristotle's classic example.
2. (a) Valid. (b) Not true: the major premise is false, so the whole argument falls apart.
3. (a) Not valid. (b) The statements may all be true, but the premises do not guarantee the conclusion: Esme could live in another place in California.

Exercise 2. (book)

1. regulate those factories, or ensure that workers are being paid fairly
2. boycott
3. clean air, or reducing air pollution

Exercise 3. (book) Any of these structures is a good answer:

- Major premise: It is better to buy ethically-made products. Minor premise: Studies show that people are not good at remembering information that shows they are making unethical choices. Conclusion: Therefore, marketers and consumers should use strategies that do not depend on people remembering bad information about products.
- Major premise: Shoppers who consider themselves moral must either (1) not buy an ethically bad product or (2) not know the product is ethically bad. Minor premise: Shoppers are unwilling or unable to stop buying ethically bad products. Conclusion: Therefore, they forget what they know about the product being ethically bad.
- Major premise: People experience discomfort when their actions do not match their values. Minor premise: Shoppers often take actions that oppose their own values, because they really want the clothing, and it is hard to find products that are ethically made. Conclusion: Therefore, they forget the information about the unethical production as a strategy to ease their discomfort.

Exercise 4.

1. That shoppers would prefer to forget when a company exploits workers or engages in other unethical actions, just as people tend to forget their own past unethical behavior ("unethical amnesia").
2. Study 1: 236 undergraduates memorized descriptions of six wooden desks (half made from rainforest wood, half from sustainable tree farms). After about 20 minutes of unrelated tasks, they were significantly less likely to remember correctly that a desk was made with rainforest wood. Study 2: 402 adults online read about jeans made by adults or by children. Those who saw the child-labor jeans were significantly less likely to remember this detail, and said they felt more uncomfortable. Study 3: 341 adults online read about a consumer who forgot whether jeans were ethically made, or one who remembered but ignored it. They judged the consumer who forgot less harshly.
3. To check whether the memory errors happened with any information about the desks or only with the ethical information. The students did not make the same kinds of errors with other attributes

such as prices, so the forgetting seems specific to the ethical information.

4. It is an interpretation. The study measured memory for the detail and how uncomfortable participants felt; that the discomfort caused the forgetting is the writers' explanation. The article does not show how they determined it, so a careful reader would want to see the full study.

My notes